

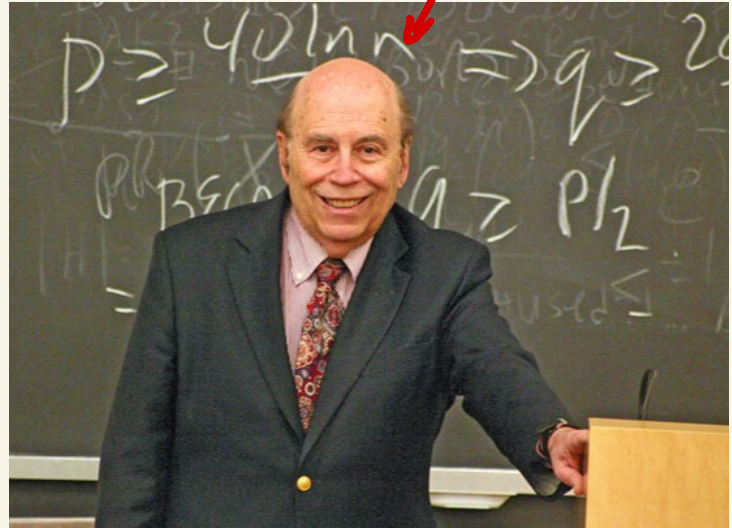
# Miller-Rabin

## Primality Test

Miller



Rabin



- Fast

- ↳ cubic running time :  $O((\log n)^3)$

- Easy

- ↳ gcd and arithmetic in  $(\mathbb{Z}_n, +, \cdot)$

- Reliable

- ↳ always correct on primes

- ↳ high probability of detecting composite

## Idea

Each composite number  $n$  has "many" numbers less than  $n$  that certify it is not prime.

| randomly sample many numbers less than  $n$   
|  $\hookrightarrow$  unlikely to get no certificates!

# Algorithm

usually a constant  $k$

Let  $p(n)$  be a polynomial in  $n$ .

TEST( $N$ )

- 1 Pick  $1 < b_1, \dots, b_{p(\log N)} < N$  uniformly random
- 2 **for**  $i = 1, \dots, p(\log N)$
- 3     Check if  $(b_i)^{N-1} \not\equiv 1 \pmod{N}$ , return NO if so
- 4     **for**  $j = e, \dots, 0$  where  $e$  is the multiplicity of 2 in  $N - 1$ 's factorization
- 5         Check if  $1 < ((b_i)^{(N-1)/2^j} - 1, N) < N$ , return NO if so
- 6 Return YES

$b < n$  is a Certificate  $\Leftrightarrow$

$b^{n-1} \not\equiv 1 \pmod{n}$ , or

$\exists i : 2^i | (n - 1) \text{ and } 1 < (b^{(n-1)/2^i} - 1, n) < n.$

# Running Time

Let  $p(n)$  be a polynomial in  $n$ .

TEST( $N$ )

- 1 Pick  $1 < b_1, \dots, b_{p(\log N)} < N$  uniformly random
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- 6 Return YES

$(\log N)^3$   
repeated squaring

$(\log N)^2$   
squaring

$$b_i^{(N-1)/2^e}$$

$$b_i^{(N-1)/2^{e-1}}$$

$\vdots$

$$j = e$$

$$j = e - 1$$

$\vdots$

$$\boxed{\log N}$$

$$\boxed{\log N}$$

$\rightarrow O((\log N)^3)$  if  $p$  constant

Theorem: At least  $3/4$  of numbers  $< n$   
are certificates if  $n$  is composite.

Proof:

too long for 10 min 😊

But ....

We can prove an easier but useful lemma

Remark: Pick a constant  $k$ , MR will detect composites with probability  $\frac{1}{2^{2k}}$ .

Lemma: Let  $m_1, \dots, m_k$  pairwise coprime and divide  $n$ . Let  $s_1, \dots, s_k$  be s.t.  $s_i < m_i$  and  $(s_i, m_i) = 1$ . Then, the number of  $b < n$  coprime to  $n$  s.t.  $b_i \equiv s_i \pmod{m_i}$  is independent of the choice of  $s_1, \dots, s_k$ .

| Let  $\bar{E}_d = \{b \in \{1, \dots, d-1\} \mid (b, d) = 1\}$ .  
|  $\hookrightarrow$  so  $s_i \in \bar{E}_{m_i}$  and  $b \in \bar{E}_n$  above.  
|  $\hookrightarrow \bar{E}_d$  is a group!

Proof: Define  $f: E_n \rightarrow E_{m_1} \times \dots \times E_{m_k}$  by

$$f(b) = (b \bmod m_1, \dots, b \bmod m_k).$$

↳ Clearly,  $f$  is a homomorphism of groups (i.e.  $f(ab) = f(a)f(b)$ )

↳ Remains to show  $f$  is onto.

| Recall that preimage of  $(s_1, \dots, s_k)$  is  
| a coset of  $E_n$  and cosets have same size

f is onto :

Let  $t$  be the product of primes dividing  $n$  but not  $m_i$  for any  $i$ .

Chinese Remainder Theorem :

$$\underline{\forall s_1 \in \bar{E}_{m_1}, \dots, s_k \in \bar{E}_{m_k}, s_{k+1} \in \bar{E}_t}$$

$$\exists b \in \bar{E}_{m_1 m_2 \dots m_k t} \quad \text{s.t.} \quad b \equiv s_i \pmod{m_i} \quad \forall i$$

and  $b \equiv s_{k+1} \pmod{t}.$



What is the lemma saying??

Given  $m_1, m_2, \dots, m_{482}$ .

Suppose  $s_1, \dots, s_{222}$  are fixed.

Given any  $s_{223}, \dots, s_{482}$ , the number of  $b$ 's in  $\bar{E}_n$  s.t.  $b_i \equiv s_i \pmod{m_i}$  is the same!

|  $k$ -wise independence

## Proof of Theorem (baby version):

Let  $n = p^k$ ,  $p$  is a prime,  $k > 1$ .

WTS at most  $1/4$  of  $\{1, \dots, n-1\}$  are not certificates.

note: | not ( 
$$\begin{array}{l} b^{n-1} \not\equiv 1 \pmod{n}, \text{ or} \\ \exists i : 2^i | (n-1) \text{ and } 1 < (b^{(n-1)/2^i} - 1, n) < n. \end{array} \quad )$$
 
$$\Rightarrow b^{n-1} \equiv 1 \pmod{n}.$$

Let  $a \in E_p$  be a primitive root and  $b < n$  be a noncertificate (so  $b^{n-1} \equiv 1 \pmod n$ ).

$$\hookrightarrow b \equiv a^k \pmod p, \text{ some } k.$$

$$\hookrightarrow b^{k(n-1)} \equiv 1 \pmod n.$$

$$\hookrightarrow \phi(n) = \phi(p^k) = p^{k-1}(p-1) \mid k(n-1)$$

$$\text{So } \frac{\phi(p^k)}{(\phi(p^k), n-1)} \mid k. \quad \leftarrow \text{explain}$$

$$\frac{\phi(p^k)}{(\phi(p^k), n-1)} = \frac{p^{k-1}(p-1)}{\underbrace{(p^{k-1}(p-1), n-1)}_{=(p-1, n-1) \text{ since } p \nmid n-1}}$$

$$\geq p^{k-1}$$

If  $n > 9$  (manually check for  $n \leq 9$ ),  
 then  $p \geq 5$  or  $k \geq 3$ . So  $p^{k-1} \geq 4$ .

This means for any  $b$  noncertificate,

$b$  must be in the form

$a^k \bmod p$ , where

$$\boxed{\frac{\phi(p^k)}{(\phi(p^k), n-1)}} \mid k$$

$\geq 4$

At most  $1/4$  of the numbers mod  $p$  have this property. Lemma implies the baby theorem.  $\square$

See

<https://github.com/Andrewfeng123/MAT482>

for an implementation in C++  
and a proof of adult version.

Thank You!